

Chapter 68. Long-Term Outcomes after Mechanical Ventilation

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Long-Term Outcomes after Mechanical Ventilation: Introduction

When the first edition of this book was published in 1994, the standard metric of outcome after critical illness was 28-day all-cause mortality. Since that time, there has been a revolution in the way we view critical illness, its treatment, and its lasting effects both for the patient and the family caregiver. The dramatic rise in the number of longer-term outcomes reports published over the past 5 to 10 years clearly reflects this point. It is now apparent that most patients who have survived an episode of severe critical illness requiring mechanical ventilation will sustain some compromise in physical function related to intensive care unit (ICU) acquired weakness and a myriad of other physical disabilities.^{1–5} This acquired disability may be permanent.⁵ Patients may also sustain an important new or incremental decline in neuropsychological function, including neurocognitive impairments, and neuropsychiatric or mood disorders.^{6–8} The constellation of muscle, nerve, and brain dysfunction^{3,9–13} may permanently alter disposition for those who were previously independent and who, post-ICU, now require assisted living or comprehensive care.¹ ICU-acquired morbidity may result in an additional cost burden in health care utilization that is similar to patients with chronic disease,^{1,5,13–14} and this traumatic life event may completely erode the reserve of family members and they may also acquire new mood disorders.^{15–18}

This chapter reviews the literature on physical and neuropsychological outcomes after critical illness and highlights important emerging data that link patient characteristics, burden of comorbid disease, and ICU practice patterns to acquisition of new morbidity. In addition, newer data on psychological disability in family caregivers and models of rehabilitation and intervention after critical illness are discussed. To date, the outcomes literature has been dominated by reports on survivors of acute respiratory distress syndrome (ARDS), but there are emerging data on other vulnerable patient subgroups, such as the frail elderly and the chronically critically ill. Outcomes in these populations add insight to the current understanding of the diversity in post-ICU disability and also are discussed. The chapter concludes with a commentary on a proposed framework for a continuum of rehabilitation for patients and families after critical illness.

Physical Morbidity after Critical Illness

The current state of the outcomes literature suggests that the greatest short-term morbidity after critical illness is diminished physical health-related quality of life as a consequence of ICU-acquired weakness.^{3,5,9,10,13,19–22} Reports from long-term follow-up studies suggest that by 5 years, patients complain less of physical disability and more of a decline in general health, diminished vitality, and mood disorders.⁵ The finding of compromised physical quality of life has been robust across many studies since the early 1990s, as discussed in detail below. An understanding, however, of the specific determinants of this reported decrease in physical function was not well known until more than a decade later.

Patients with acute lung injury and ARDS represent some of the most complex critically ill and long-stay ICU patients and have been the most rigorously studied group of ICU survivors to date. More than 100,000 patients survive acute lung injury and ARDS every year²⁴ and their outcomes dominate the review that follows. The emerging outcomes literature on the chronically critically ill, elderly, and sepsis patient populations adds nuance and depth to our understanding of the spectrum of disability and is also included.

Physical Function and Health-Related Quality of Life

Health-related quality of life is a set of causally linked dimensions of health, including biologic/physiologic, mental, physical, social function, neurocognitive, and health perception.²¹ Measures of health-related quality of life have emerged as important patient-centered metrics of recovery from critical illness. These questionnaires provide ease of administration and ready interpretability with published normative data for comparison, although fail to yield crucial details about the exact nature of any reported disability. Without this insight or more detailed understanding, the investigator is left to blindly hypothesize about the etiology of the reported disability and limits her or his ability to construct appropriate interventional or rehabilitation programs.

There is emerging evidence that the degree of disability acquired after critical illness and resultant health-related quality of life is variable and relates to differences in premorbid functional status, burden of comorbid illness, and nature and duration of critical illness.^{1,3,4} Although there is some heterogeneity across different study samples of patients with ARDS, there is less variability in reported health-related quality of life among this group than in general populations of critically ill patients.²⁴ The following is an overview of the literature on ARDS health-related quality of life and these data highlight the consistent signal of diminished physical function reported up to 1 year after severe lung injury, and also the controversy that still continues about the relative contribution of pulmonary disability to functional decline after ARDS and severe critical illness. In 1994, McHugh et al were the first group to prospectively evaluate the relationship between pulmonary dysfunction and functional disability in patients with ARDS.¹⁹ They noted that scores for the Sickness Impact Profile (a generic quality-of-life measure of the subject's self-perceived physical and psychological condition) rose substantially to 3 months postextubation with only minimal further improvement to 1 year. Using a lung-related Sickness Impact Profile score, only a modest proportion of the patients' overall physical dysfunction was attributed to pulmonary difficulties. Weinert et al²⁰ documented functional impairment in a cohort of lung injury survivors using the Medical Outcomes Study 36-item short-form health survey (SF-36) and reported the

largest decrements in role-physical and physical-functioning domains, which patients attributed to global disability.²⁵ Similar observations were made by other groups who inferred that disability in their patients with ARDS was secondary to pulmonary dysfunction in the absence of any direct pulmonary measurement²⁶ or through administration of pulmonary disease-specific measures and this failed to differentiate between pulmonary and extrapulmonary disability.²² Others have made comparisons to control populations of patients with known pulmonary disease (e.g., cystic fibrosis), but after health-related quality-of-life measures had been disaggregated, symptom burden was seen to be related to musculoskeletal or constitutional issues.¹⁰ In the context of outcome evaluation after a trial comparing higher versus lower tidal volumes during mechanical ventilation, some have demonstrated very modest abnormalities of pulmonary function, which correlate with reduced function on health-related quality-of-life metrics, and maintain that there is an important relationship between pulmonary and functional outcomes in ARDS survivors.²⁸

In 2003, Herridge et al reported on 109 ARDS survivors at 3, 6, and 12 months after ICU discharge and captured function as distance walked in 6 minutes, pulmonary function, and health-related quality of life using the SF-36. The investigators also did a detailed in-person interview and physical assessment as part of each follow-up visit. In their relatively young patient sample (median age: 45 years), with few comorbidities and who had been working full time before their critical illness, 6-minute walking distance was 66% at 1 year and pulmonary function was normal or near normal. They noted that the major morbidity after ARDS and sepsis was muscle wasting and weakness.³ In-person evaluation of patients and the collection of multiple outcome measures led the investigators to different conclusions about the major contributors to morbidity after severe lung injury and also facilitated the generation of a catalogue of physical disabilities that contribute to poor health-related quality of life (Fig. 68-1).

Figure 68-1

Heterotopic ossification



Alopecia

Tracheal stenosis

Cosmesis-Scars from CVC, Art line, CT, drain sites

Hearing loss

Pulmonary fibrosis

Frozen joints contractures

Renal impairment

Nerve and muscle



Weakness

Brain



Mental health & cognition

Bronchiectasis

Taste changes

Ischemic digits

Striae



Spectrum of disability

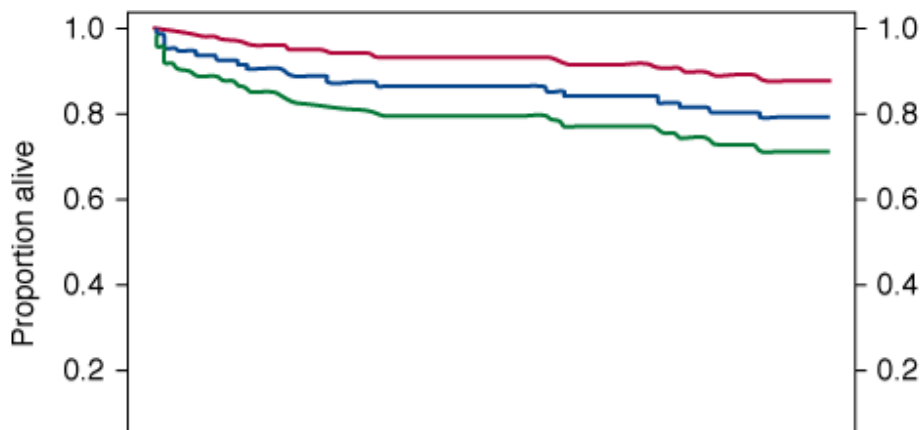
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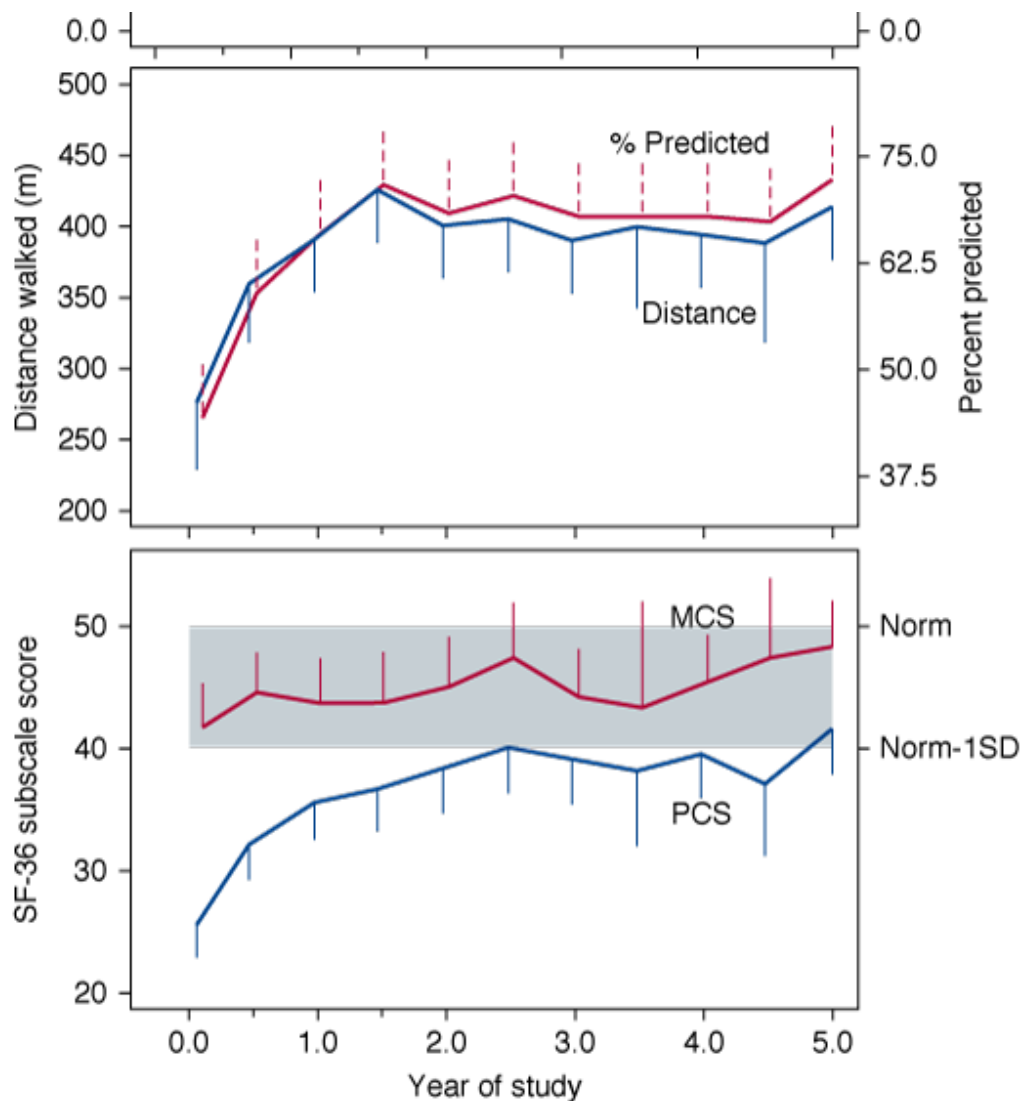
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The spectrum of disability in survivors of critical illness. Brain, muscle, and nerve are the major morbidities; others are less common.

Impaired physical function is a robust observation across different studies, countries, and investigators, and may persist for years after ICU or hospital discharge and be irreversible (Fig. 68-2).^{4,5} Health care costs generated by relatively young, previously working patients in an ARDS cohort followed to 5 years were higher than predicted for this age group and comparable to patients with chronic disease (Fig. 68-3). It was not expected that this patient group would not fully recover from their critical illness in terms of physical functioning, emotional outcomes, or quality of life, and this finding has significant public health ramifications.

Figure 68-2





N at risk	109	92	86	79	77	74	69	64
N for 6 min walk	80	78	81	60	64	64	57	54
N for SF-36	67	74	74	56	57	57	49	50

- Exact survival times were used for these analyses whereas deaths indicated in the consort diagram were included between scheduled follow-up visits.

Top Panel - Kaplan-Meier curve to 5 years. Dashed lines represent the 95% confidence interval

Middle Panel - Distance walked in 6 minutes (meters and % predicted); Distance in meters is a solid line; and % predicted is a dashed line.

Bottom Panel - SF-36 Subscale scores for Physical Component Score (PCS) and Mental Component Score (MCS)

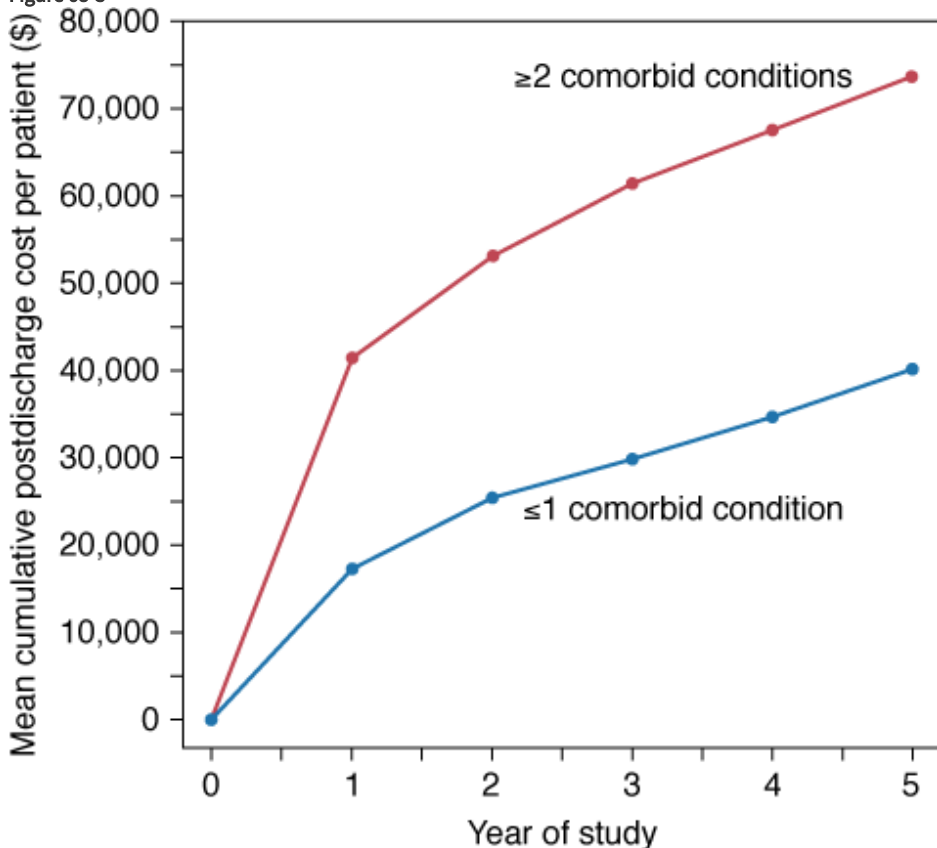
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Top Panel: Survival—Kaplan-Meier curve to 5 years. *Dashed lines* represent the 95% confidence interval. Exact survival times were used for these analyses. *Middle Panel:* Six-minute walk distance—Distance walked in 6 minutes (meters and % predicted); distance in meters is a *solid line* and % predicted is a *dashed line*. *Bottom*

Panel: Quality of life to 5 years after ICU discharge—SF-36 Subscale scores for Physical Component Score (PCS) and Mental Component Score (MCS). (Used, with permission, from Herridge et al.⁵)

Figure 68-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Cumulative costs to 5 years after ICU discharge stratified by number of comorbid conditions at ICU admission.

Lower line is less than or equal to one comorbid condition and *upper line* is greater than or equal to two comorbid conditions. (Used, with permission, from Herridge et al. *N Engl J Med*. 2011.⁵)

Risk Factors for Poor Functional Outcome after Critical Illness

The ARDS outcomes data are instructive because most of these study samples involved younger patient groups with few or no premorbid conditions. It was a surprising and striking observation that these patients did not fully recover and had important and persistent morbidity for years after their critical illness. These data establish that there is a physical and neuropsychological “cost” to a severe episode of critical illness, and that most patients, particularly those in more vulnerable groups, such as patients requiring prolonged mechanical ventilation, patients with a significant burden of comorbid illness, and the elderly, may suffer even greater functional decline and require dependent care as a result of their critical illness.

Studies on survivors of prolonged mechanical ventilation have helped in elucidating important risk factors for outcome and change in disposition. Chelluri et al evaluated factors associated with mortality and quality of life in 817 patients 1 year after prolonged mechanical ventilation.²⁸ The median age of their patients was 65 and 1-year survival was 44%. Those patients surviving their ICU stay had fewer comorbidities, lower severity of illness

scores, and less premorbid dependence in activities of daily living. Fifty-seven percent of surviving patients needed caregiver assistance at 1 year of follow-up. A French study by Combes et al evaluated 347 patients receiving mechanical ventilation for equal to or greater than 14 days.²⁹ Factors associated with death in the ICU included age greater than 65 years, preadmission New York Heart Association functional class of equal to or greater than 3, preadmission immunocompromised status, septic shock at ICU admission, need for renal replacement therapy in the ICU, and nosocomial septicemia.

Unroe et al evaluated the trajectories of care and resource utilization for 126 patients with a median age of 55 years who also required prolonged mechanical ventilation. These patients had, on average, two comorbid conditions, and most were not working, retired, or disabled at the time of ICU admission. At 1 year, only eleven patients (9% of the cohort) were alive and without functional dependency. Patients with poor outcomes were older, had more comorbid conditions, and were more frequently discharged to a post-acute care facility. The mean cost per patient was \$306,135 (standard deviation [SD]: \$285,467) and total cohort cost was \$38.1 million, for an estimated \$3.5 million per independently functioning survivor at 1 year.¹

Iwashyna et al noted persistent reduction in functional status after sepsis and critical illness, again echoing the theme of acquired disability after critical illness. Among older patients (median age: 77 years), they observed a high rate of new functional limitations in those who had no limits before their episode of sepsis (mean: 1.57; 95% confidence interval [CI]: 0.99 to 2.15). In patients with reductions in activities of daily living before sepsis, they noted an important further decrement in function. Neurocognitive and physical decline persisted for at least 8 years after the episode of sepsis and altered patients' ability to live independently.⁴

The impact of increased age, however, on outcomes after critical illness remains somewhat controversial. Some studies report higher mortality with advanced age³⁰ and others do not.^{31,32} Khouli et al³³ recently evaluated 484 patients, who were age 65 years and older, and administered an health-related quality-of-life instrument to the patient or proxy at ICU admission and 6 months after hospital discharge. One-third of these patients died within 6 months of hospital discharge. Independent predictors of death at 6 months were: number of days during the 30 days before hospitalization that the patient felt their "physical health was not good," a higher Acute Physiology and Chronic Health Evaluation (APACHE) II score, and chronic pulmonary disease. This same group also found that the oldest survivors (age 86 years) had worse health-related quality of life over time, including more days spent with poor physical and mental health, compared to baseline. There appears to be a clear prognostic signal in the elderly in terms of physiologic reserve, burden of chronic organ dysfunction, and nature of the health trajectory before the critical illness with regards to prognostication for survival, function, and health-related quality of life; emphasis on physiologic rather than chronologic age may provide valuable insight into projected outcome.

Neuromuscular Dysfunction

Muscle weakness and impaired function constitute an important morbidity of severe critical illness. A continuum of weakness begins within hours of mechanical ventilation,³⁴ is demonstrable by bedside evaluation within 1 week of ICU admission using the Medical Research Council scoring system,³⁵ and may

persist with incomplete recovery for years after ICU discharge^{5,36–41} (see Fig. 68-2). In the ICU context, muscle weakness has been linked to prolonged mechanical ventilation,^{42,43} delay in ICU and hospital discharge, increase in associated costs,^{44,45} and increased likelihood of death.⁴⁶

Many challenges confound the ability to characterize and intervene in neuromuscular disability after critical illness. Important heterogeneity exists in susceptibility and risk factors across different critically ill patient populations. There are no validated measures to reliably risk stratify patients according to physical disability and no consistent nomenclature for the spectrum of lesions that constitute muscle weakness and wasting. Currently, different terminologies coexist in the literature and include critical illness polyneuropathy, critical illness polyneuropathy and myopathy, critical illness neuromyopathy, ICU-acquired paresis, critical illness myopathy and/or neuropathy. This complexity is compounded by the observation that nerve and muscle injury may coexist, and it is difficult to delineate discrete risk factors and natural history. There are obstacles in terms of the variety and sensitivity of testing methods, criteria for diagnosis, surveillance and selection bias in evaluation and reporting. Sensory and motor evaluations are limited in heavily sedated patients, and, therefore, clinical bedside testing and determination of the prevalence of these lesions may be unreliable.

Despite these limitations, a recent classification system was proposed by Stevens et al (Table 68-1).⁴⁷ This system is simple, accessible, and clinically relevant, and emphasizes clinical assessment and bedside testing. It proposes a broad classification for ICU-acquired weakness and highlights it as a diagnosis of exclusion, onset during the critical illness, ventilator dependence, and weakness demonstrable by use of bedside hands-on muscle testing (Medical Research Council score). A diagnosis of critical illness polyneuropathy is made in those who meet criteria for ICU-acquired weakness and have electrophysiologic evidence of a sensorimotor axonal polyneuropathy. A diagnosis of pure critical illness myopathy is made in patients who meet criteria for ICU-acquired weakness and who have myopathic features on electromyography during muscle contraction and/or myopathic biopsy.

Table 68-1: Summary of Diagnostic Criteria for Intensive Care Unit-Acquired Weakness, Critical Illness Polyneuropathy, and Critical Illness Myopathy

Diagnostic criteria for ICU-acquired weakness (ICUAW) 1, 2, 3 or 4, 5 below

1. Generalized weakness developing after the onset of critical illness
2. Weakness is diffuse (involving both proximal and distal muscles), symmetric, flaccid, and generally spares cranial nerves
3. MRC (Medical Research Council) sum score^a <48, or mean MRC score <4 in all testable muscle groups noted on two or more occasions separated by >24 hours
4. Patient dependent on mechanical ventilation
5. Other causes of weakness not related to underlying critical illness have been excluded

Diagnostic criteria for critical illness polyneuropathy (CIP)

1. Patient meets criteria for ICUAW
2. Compound muscle action potential amplitudes are decreased to <80% of lower limit of normal in two or more nerves
3. Sensory nerve action potential amplitudes are decreased to <80% of lower limit of normal in 2 or more nerves
4. Normal or near-normal nerve conduction velocities without conduction block
5. Absence of a decremental response on repetitive nerve stimulation

Diagnostic criteria for critical illness myopathy (CIM)

1. Patient meet criteria for ICUAW
2. Sensory nerve action potential amplitudes are >80% of the lower limit of normal in two or more nerves
3. Needle electromyogram in two or more muscle groups demonstrates short-duration, low-amplitude motor unit potentials with early or normal full recruitment with or without fibrillation potentials
4. Direct muscle stimulation demonstrates reduced excitability (muscle-to-nerve ratio >0.5 in two or more muscle groups)
5. Muscle histology is consistent with a myopathy

Probable CIM: criteria 1, 2, 3, or 4; or 1 and 5 (listed above)

Definite CIM: criteria 1, 2, 3 or 4, 5 (listed above)

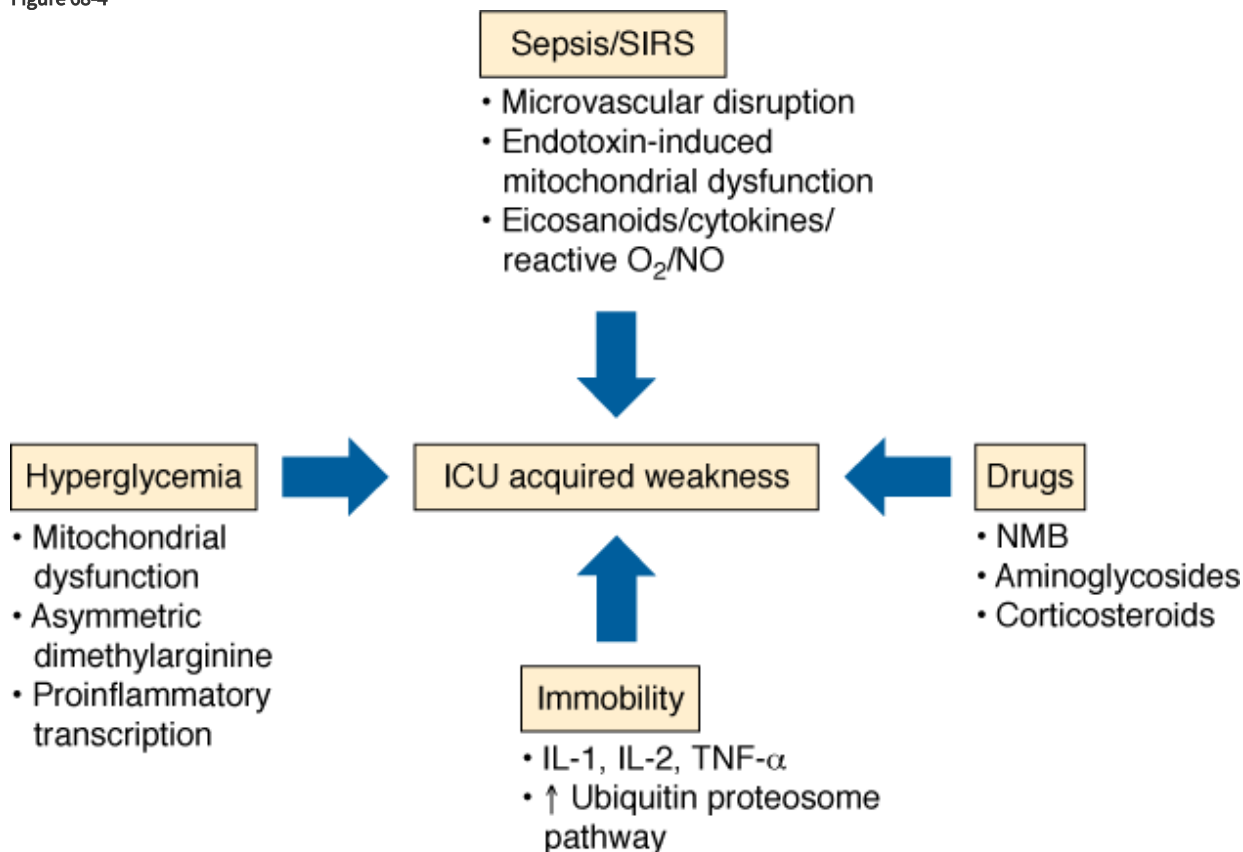
^aMRC Sum score assesses three muscle groups in each of the upper and lower limbs. Each muscle group score ranges from 0 (paralysis) to 5 (normal muscle strength), and the overall score ranges from 0 to 60.

Source: Adapted, with permission, from Stevens RD, Marshall SA, Cornblath DR, et al. A framework for diagnosing and classifying intensive care unit-acquired weakness. *Crit Care Med*. 2009;37(10 Suppl):S299–S308.

This section discusses the pathophysiologic mechanisms for muscle and nerve injuries (Fig. 68-4). Given the limitations in this distinction, as outlined above, the mechanisms are broadly grouped as critical illness

polyneuropathy and critical illness myopathy.

Figure 68-4



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Risk factors for ICU acquired weakness. *IL-1*, interleukin-1; *IL-2*, interleukin-2; *NMB*, neuromuscular blockers; *NO*, nitric oxide; *TNF- α* , tumor necrosis factor alpha. (Adapted, with permission, from Stevens RD. *Crit Care Clin*. 2008;24:1–24.)

Critical Illness Polyneuropathy

Background and Incidence

Bolton et al published a landmark study on critical illness polyneuropathy in 1984.⁴⁸ They described a primary axonopathy that presented as a mixed sensorimotor neuropathy in five critically ill patients who were ventilator dependent. Detection of the true incidence of critical illness myopathy is complicated by poor consensus on surveillance, timing, and nature of testing, and formal definition and diagnostic criteria (as outlined above). When patients are evaluated solely on the presence of clinical weakness, studies demonstrate an incidence of 25% to 36%.^{35,49} A review of 1421 critically ill patients, evaluated using diagnostic tests (nerve conduction velocities, needle electromyography, direct muscle stimulation, histopathology of muscle or nerve tissue) or a combination of test findings and clinical findings, reported an incidence of critical illness neuromyopathy of 46% (95% CI: 43% to 49%).⁵⁰

Etiology and Pathophysiology

Systemic Inflammatory Response Syndrome and Sepsis

Critical illness polyneuropathy has been associated with sepsis and systemic inflammatory response syndrome in multiple cohort studies.⁵⁰ In sepsis, critical illness polyneuropathy is linked to microcirculatory perturbations with resultant axonal injury and degeneration. A recent report describes proinflammatory cytokine (tumor necrosis factor-(and interleukin-1)-mediated increased expression of E-selectin on the endoneurial and epineurial vessels of peripheral nerves.⁵¹ There may be functional disruption of nerve action potential early in the course of disease, which may be a harbinger of later structural derangement of the nerve.⁵²

Hyperglycemia

Hyperglycemia in critically ill surgical and medical populations is consistently associated with ICU-acquired weakness.^{53,54} Van den Berghe et al demonstrated that tight glycemic control reduced critical illness polyneuropathy as defined by neurophysiologic testing, from 51.9% in control subjects to 28.7% among insulin-treated patients.^{53,54} Hyperglycemia may be harmful based on mitochondrial dysfunction and the deleterious effects of oxidant injury and apoptosis.^{55,56} Derangement of **nitric oxide** production is also a postulated mechanism,⁵⁷ as is the role of **insulin** in inhibiting proinflammatory transcription factors and promotion of neuroregeneration during critical illness.^{58,59}

Pharmacologic Agents

Early reports suggested an association between neuromuscular dysfunction and exposure to neuromuscular blockers and systemic corticosteroids.^{60–62} This risk was profiled in patients with status asthmaticus who developed ICU-acquired weakness and received treatment with both agents.⁶³ These risks have not been borne out in systematic review because of confounding by exposure to aminoglycosides, vasopressors, and renal replacement therapy in sepsis and systemic inflammatory response syndrome patients.^{50,64–65} In a 1-year outcomes study in patients with ARDS, exposure to any systemic corticosteroid was significantly associated with a decrease in 6-minute walking distance at 3 months in a multivariable model, and patients demonstrated important proximal weakness in the shoulder and hip girdle at that time point.³ A recent trial using 48 hours of continuous paralytic in patients with severe lung injury saw significant organ function improvement, and the continuous paralytic was not associated with an increased risk of ICU-acquired weakness assessed by manual muscle testing at ICU discharge.⁶⁶ No long-term follow-up, however, was done, and there is concern that manual muscle testing may not have adequate sensitivity to detect ICU-acquired weakness.⁶⁶ Early corticosteroid use appears to be strongly associated with ICU-acquired weakness, although the risk associated with paralytic use remains somewhat uncertain.

Critical Illness Myopathy

Background and Incidence

Critical illness myopathy currently encompasses critical illness myopathy, acute quadriplegic myopathy, thick filament myopathy, and necrotizing myopathy, and the incidence varies between 48% and 96% in studies that

included muscle biopsy.⁶⁵ Critical illness myopathy is a nonnecrotizing diffuse myopathy associated with fatty degeneration of muscle fibers, fiber atrophy, and fibrosis.⁶⁷ This lesion has also been linked to corticosteroid and paralytic exposure and may be clinically indistinguishable from critical illness polyneuropathy because patients are also weak, paretic, and difficult to wean.

Thick-filament myopathy shows a selective loss of myosin filaments in the context of significant corticosteroid or neuromuscular blocker exposure and immobility.⁶⁸ Acute necrotizing myopathy is distinguished by extensive myonecrosis with vacuolization and phagocytosis of muscle fibers, and is linked to corticosteroid and neuromuscular blocker exposure and multisystem organ dysfunction.⁶⁹

Etiology and Pathophysiology

The pathophysiology of critical illness myopathy involves catabolism, inflammation, and derangement of membrane excitability. An increase in urinary nitrogen loss, low glutamine, protein, and DNA levels in muscle biopsies, and upregulation of the calpain and ubiquitin proteolytic pathways and apoptosis have all been documented.⁷⁰

Inactivity in critically ill patients propagates inflammatory mediators, which stimulate protein loss in differentiated muscle cells, activate signaling events that promote oxidative injury and disruption of insulin receptor signaling with resultant substrate reduction and impairment of myofibril growth and repair.⁷¹ Interleukin-1, interleukin-6, and tumor necrosis factor- α have proinflammatory properties and have all been implicated in muscle degradation in critical illness and augmentation of proteolysis and muscle loss. Muscle membrane inexcitability may also contribute to weakness and is related to inactivation of sodium channels. Allen et al recently reported altered muscle-fiber excitability and evidence for muscle membrane dysfunction as the principal underlying abnormality in critical illness myopathy.⁷²

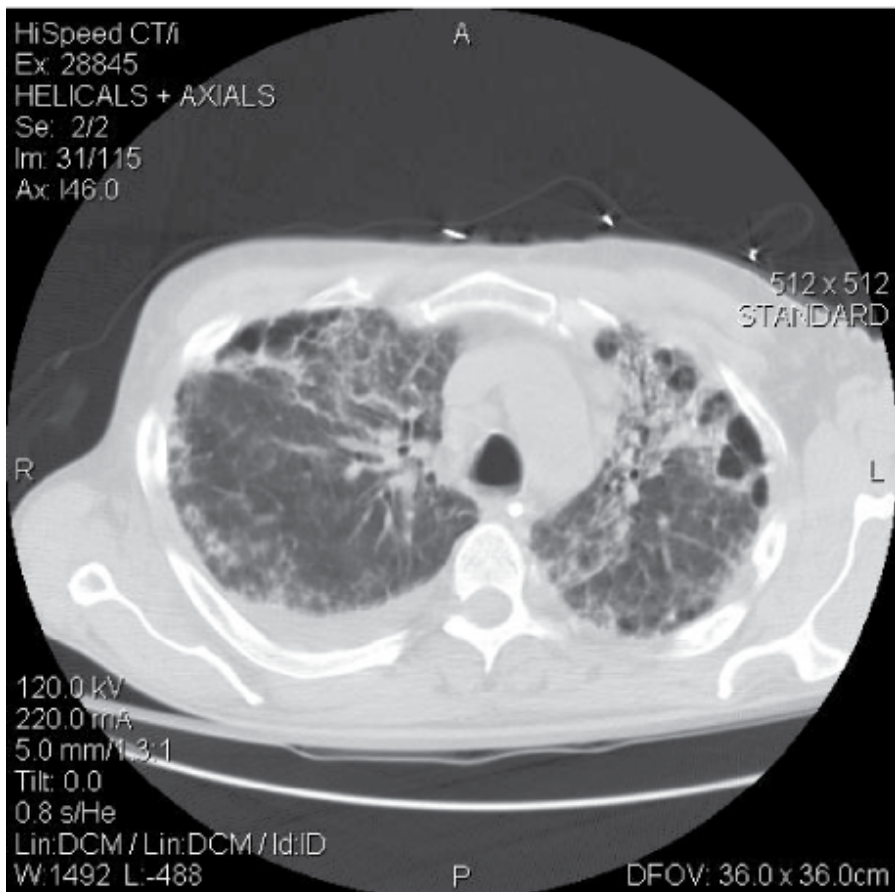
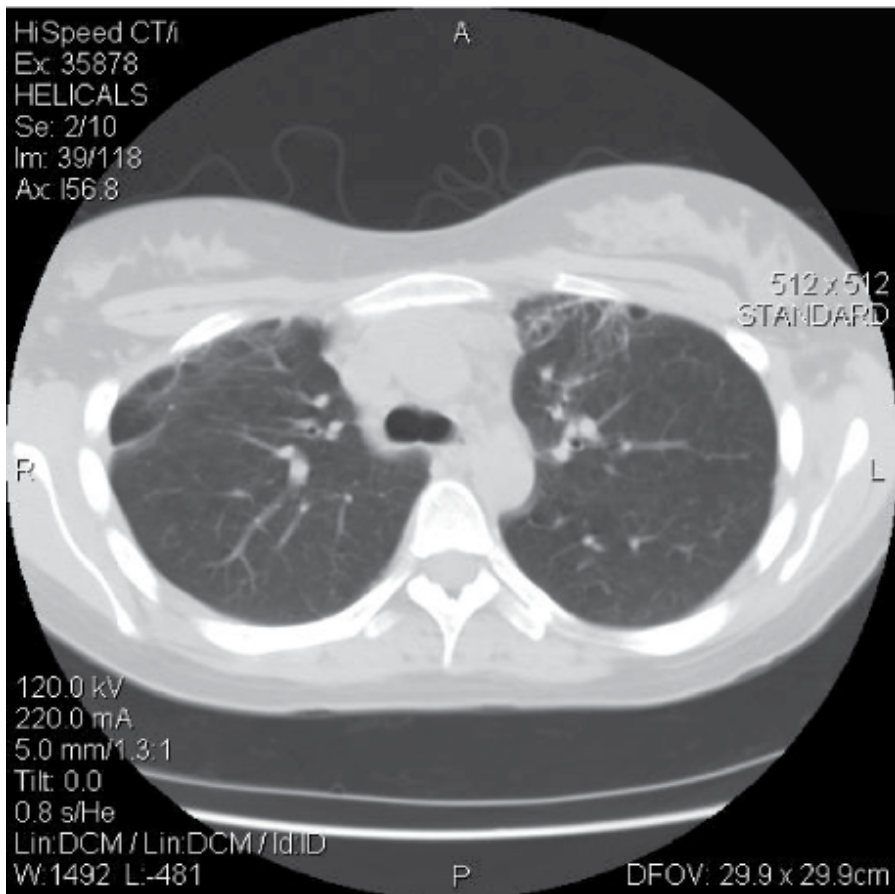
Pulmonary Function

Pulmonary function outcomes may be heterogeneous after an episode of ARDS. Many ARDS survivors have persistent pulmonary function impairments: typically, mild restrictive changes with an associated reduction in diffusion capacity.^{3,19} Others have documented more diversity in outcome, with significant variability in the proportion of patients with obstructive (0% to 33%) and restrictive (0% to 50%) defects, as well as compromised diffusion capacity (33% to 82%).⁷³ There are reports of mild pulmonary function abnormalities associated with decreased health-related quality of life 1 year following hospital discharge²⁷ and no improvement in pulmonary function after the first year.⁷⁴ Most recent data from a 5-year outcomes cohort study in patients with severe ARDS shows normal to near normal pulmonary function achieved by 6 months to 1 year after ICU discharge and continued stability over the 5-year study period.⁵

There are imaging studies of ARDS survivors at 6 to 10 months,⁷⁵ 3 years,⁷⁶ and 5 years.^{5,77} Investigators have noted that most patients with ARDS had minor localized changes in the nondependent lung zones (Fig. 68-5),

and were able to document some association with severity of lung injury and duration of mechanical ventilation.^{75,76} There were also findings of bronchiectasis or new pulmonary fibrosis in one-third of patients in one study⁵ (Fig. 68-5) and loss of ventral distribution in reticular pattern in patients treated with extracorporeal membrane oxygenation.⁷⁷

Figure 68-5



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Top Panel: Computed tomography (CT) findings at 1 year after acute respiratory distress syndrome (ARDS) in a young, previously healthy person without preexisting lung disease. Finding of minor nondependent fibrosis and minor cystic changes were most common in this group. *Bottom panel:* CT findings at 1 year after ARDS in same group of ARDS survivors with an example of a rare finding of more marked nondependent fibrosis, bronchiectasis, and cystic changes.

The variability in reported pulmonary outcomes to date suggests possible confounding by smoking history, preexisting obstructive or restrictive pulmonary disease, physiologic restriction related to ICU-acquired weakness affecting respiratory muscles, presence of other pulmonary processes that fulfill the ARDS definition but that have a very different natural history (e.g., cryptogenic organizing pneumonia), and loss to follow-up that continues to challenge validity and interpretation of follow-up data. Most outcome studies found ARDS survivors are often unable to resume their prior physical function, and the degree of pulmonary dysfunction documented across studies does not solely explain their degree of functional limitation.

Additional Physical Morbidities

The spectrum of physical morbidities after critical illness is shown in [Figure 68-1](#).

Entrapment Neuropathy

The Toronto ARDS Outcomes study observed a 6% prevalence of peroneal and ulnar nerve palsies.³ Although this represents only a small proportion of patients, these nerve palsies complicated rehabilitation therapy and precluded return to original work in some cases. Recovery was protracted, although by 5 years all had resolved.

Heterotopic Ossification

Heterotopic ossification is the deposition of paraarticular ectopic bone and has been previously associated with polytrauma, burns, pancreatitis, and ARDS.⁷⁸ Heterotopic ossification is linked with paralysis and prolonged immobilization. There was a 5% prevalence of heterotopic ossification in the Toronto ARDS cohort study with all patients having large joint immobilization, leading to important functional limitation.³ Heterotopic ossification is remediable with appropriate surgical intervention, and screening for it may help to improve long-term functional outcomes (see [Fig. 68-1](#)).

Cosmesis

After a life-threatening event, concerns about cosmesis might be considered trivial. The physically transformative nature of critical illness, however, cannot be overstated. Many patients suffer from the often devastating emotional effects related to their altered appearance. From the 5-year ARDS outcomes study by Herridge et al,⁵ patients had ongoing concerns about cosmesis, including scars from laparotomy; chest tube; central line, arterial line, and tracheostomy insertion; burns; striae from volume overload (see [Fig. 68-1](#)); and facial scars from prolonged noninvasive mask ventilation. Many patients underwent tracheostomy revision. Patients reported that cosmetic concerns contributed to social isolation and sexual dysfunction.

Critical Illness and Brain Injury

Psychiatric Morbidity

Psychiatric disorders are common among survivors of critical illness. The prevalence of depression and anxiety in different samples of ICU survivors ranges from 10% to 58%.^{12,79–82} In a recent systematic summary, Davydow et al reported that 28% of post-ICU patients had clinically significant depression.⁸¹ Neither sex, age, nor severity of illness at ICU admission were consistent risk factors for this, but early post-ICU depressive symptoms were a strong risk factor for subsequent depressive symptoms and associated with a reduction in quality of life.⁸²

Depression in ARDS survivors ranges from 17% to 58%, with some suggestion that patients with ARDS have a greater degree of depression compared to populations of general critically ill patients.^{8,20,79,81,82} Risk factors associated with depression in patients with ARDS have included duration of mechanical ventilation, ICU length of stay and sedation, alcohol dependence, female gender, and younger age.^{8,82} Predictors of anxiety at 1 year included ratio of arterial oxygen tension to inspired oxygen fraction and duration of mechanical ventilation; prediction of anxiety at 2 years was anxiety at 1 year.⁸² A recent study found that hypoglycemia may be an important risk factor for depression in ARDS survivors and this warrants further investigation.⁸³ Post-ICU affective disorders are likely multifactorial and further study will be needed to better understand patient predisposition, illness, and treatment-specific determinants of affective morbidity, and appropriate tools for diagnosis and monitoring.

Posttraumatic stress disorder (PTSD) is the development of characteristic symptoms that occur following a traumatic event(s) where triggers include a serious personal threat experienced with helplessness and intense fear.⁸⁴ The diagnostic criteria include a history of traumatic event(s) accompanied by symptoms from each of three symptom clusters: hyperarousal symptoms, intrusive recollections, and avoidance or numbing symptoms. A number of studies have examined relationships between life-threatening critical illnesses and its treatment and the development of PTSD. Schelling et al published a landmark paper on PTSD after critical illness.²⁶ These authors evaluated health-related quality of life and PTSD in a cohort of eighty ARDS survivors 4 years after ICU discharge. Almost one-third of the ARDS survivors reported impaired memory, bad dreams, anxiety, and sleeping difficulties after ICU discharge. The prevalence of PTSD was 28% and the number of adverse ICU-related memories recalled by patients was the major risk factor for PTSD. Kapfhammer et al found that 44% of critically ill patients developed PTSD at hospital discharge and 24% had PTSD symptoms 8 or more years later.⁸⁵ Davydow et al reported the median point prevalence of questionnaire-ascertained “clinically significant” PTSD symptoms was 22%, and the median point prevalence of clinician-diagnosed PTSD was 19% in populations of general critically ill patients.⁸⁶ Prior psychopathology, greater ICU benzodiazepine administration, post-ICU memories of frightening and/or psychotic experiences in the ICU, and absence of any ICU memory were consistent risk factors for post-ICU PTSD.⁸⁷

Patients with ARDS may be more susceptible to PTSD than are critically ill patients in general, and these symptoms may persist for years. The prevalence of PTSD, as diagnosed by a psychiatrist, has been reported to be 44%, 25%, and 24% at hospital discharge, 5 years and then 8 years later, respectively.⁸

The etiology of psychiatric disorders following critical illness may be secondary to sequelae of brain injury sustained from critical illness and/or its treatments, a psychological reaction to the emotional and physiologic stress of critical illness, or both. Factors such as medications, physiologic changes, pain, altered sensory inputs, and an unfamiliar environment are all potential contributors to the development of psychiatric sequelae.^{88–90} A recent review article found an association between recall of delusional memories after ICU discharge and PTSD-related symptoms, depression, and anxiety.⁹¹ Factual memories do not seem to protect survivors from experiencing symptoms of PTSD. A study by Myhren et al, which evaluated 194 patients, found that 27% had symptoms of posttraumatic stress and predictors of PTSD were higher education level, optimism, factual recall, and memory of pain.⁹²

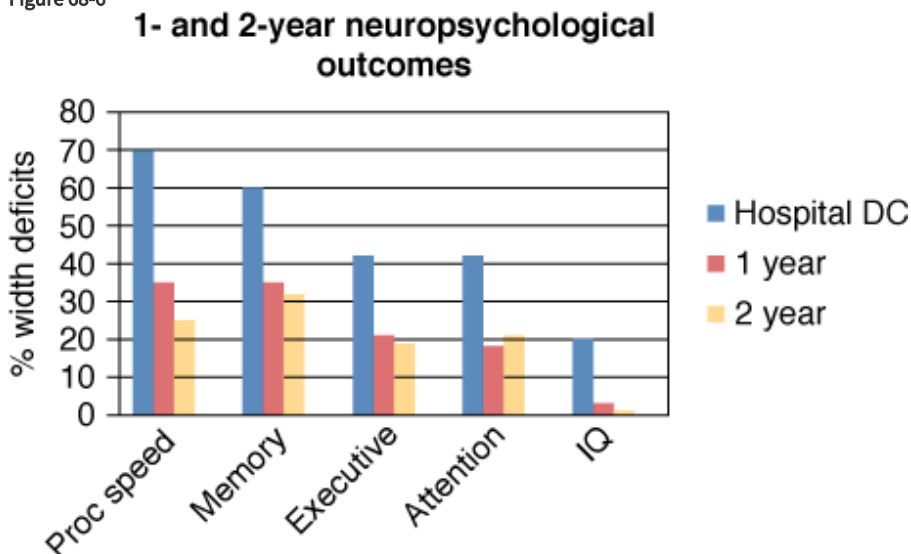
Although we are just beginning to appreciate how long-standing and debilitating psychiatric disorders are following critical illness and the important contribution they make to decreased health-related quality-of-life and functional outcomes, recent studies are beginning to investigate potential interventions to prevent or reduce psychiatric sequelae. A recent review paper suggests that corticosteroid administration may be protective for post-ICU PTSD.⁹³ A novel study suggests that use of ICU diaries in critically ill patients may reduce the incidence of PTSD. Jones et al conducted a randomized, controlled trial, in which patients were provided with an ICU diary that contained information and photographs from their ICU stays.⁹⁴ Of the patients who received the diary, only 5% had clinically significant PTSD symptoms as compared to 13% of controls. Furthermore, the patients who experienced the greatest benefit from the ICU diary intervention were those who had substantial early PTSD symptoms.⁹⁴

Neurocognitive Impairments

Critical illness is associated with new neurocognitive impairments^{7,11,12} that may be permanent. In their landmark paper in 1999, Hopkins et al were the first to demonstrate that all survivors in their ARDS cohort had cognitive impairment, including deficits in memory, attention, or concentration, and a global loss of cognitive function at hospital discharge and continued dysfunction in 30% of patients at 1-year follow-up.¹¹ These findings persisted at 2-year follow-up (Fig. 68-6).¹² These observations are robust and other investigators have documented the persistence of neurocognitive impairments in large numbers of patients at 2 months,⁹ 9 months,⁹⁵ 1 year,¹¹ 2 years,¹² and 6 years.⁹⁶ Neurocognitive dysfunction appears to improve during the first 6 to 12 months after hospital discharge. The neurocognitive impairments are often long lasting and quite severe, and many patients continue to experience significant chronic neurocognitive impairments years after ICU discharge.⁹⁶ To date, almost 1000 patients in fifteen cohorts^{79,96–100} have had some form of neurocognitive assessment after critical illness. Risk factors have included the nature of the insults experienced during critical illness and its treatment, presence of preexisting neurologic abnormalities, older age, and comorbid disorders

that might render specific domains more vulnerable to critical illness-induced brain injury. For example, patients with ARDS with neurocognitive sequelae all fell below the sixth percentile of the normal distribution of neurocognitive functioning, and had significant deficits in wide-ranging cognitive domains, including memory, executive functioning, and mental-processing abilities.²

Figure 68-6



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Domains of persistent neurocognitive dysfunction 1 to 2 years after an episode of acute respiratory distress syndrome. These include reductions in processing speed, memory, executive function, attention and concentration, and IQ. (Adapted, with permission, from Hopkins et al.¹¹ *J Int Neuropsych Assoc*. 2003;9:584.)

The question remains whether critical illness and/or its treatment is the cause of observed neurocognitive impairments or if the impairments disclose preexisting subclinical brain dysfunction. A recent longitudinal cohort study in older adults without premorbid neurocognitive impairments or dementia assessed neurocognitive function before and after an acute care or ICU hospitalization.¹⁰¹ Those who underwent acute care or critical illness hospitalization had a greater decline in neurocognitive function and new incident dementia compared to individuals who were not hospitalized. The Health and Retirement Study findings were consistent with this observation. That study followed more than 27,000 older Americans who had neurocognitive assessment both before and after sepsis.⁴ Patients with severe sepsis developed new, important, and persistent neurocognitive impairment. Thus, factors associated with acute or critical illness may be causally related to neurocognitive decline in older critically ill patients.^{4,101}

Pathophysiologic Mechanisms of Neurocognitive Impairments

The etiology of neurocognitive dysfunction is complex, multifactorial, and interacts with concurrent comorbidities and individual susceptibility. It cannot be explained by severity of illness and, specifically, is not related to severity of illness scores, medical data, age, smoking, alcohol abuse, ICU length of stay, duration of mechanical ventilation, tidal volume, or days receiving sedative, narcotic, or paralytic medications.¹²

Candidate pathophysiologic mechanisms may include hypoxemia,¹¹ sedatives or analgesics,¹⁰² hypotension,⁹⁹ delirium,¹⁰³ hyperglycemia,⁸³ and sepsis and inflammation.¹⁰⁴

Hypoxemia

Hopkins et al evaluated pulse oximetry in a prospective cohort of mechanically ventilated ARDS survivors and studied the relationship between the duration and severity of a mean oxygen saturation below 90 and neurocognitive outcome.¹¹ Mean saturations were below 90% for 122 ± 144 hours per patient, and the extent of hypoxemia correlated significantly with neurocognitive dysfunction (r^2 0.25 to 0.45; all $p < 0.01$).¹¹ Hypoxia may be associated with cortical atrophy^{103–107} and ventricular enlargement consistent with neuronal cell loss, as well as an increase in ventricular volume.^{106–108}

Hypotension

The relative contribution of hypotension to poor neurocognitive outcomes may be more equivocal. Previous studies have demonstrated a correlation between mean blood pressure less than 50 mm Hg and memory scores at hospital discharge.¹⁰⁹ Others have reported that the duration of hypotension only modestly correlated with impaired memory at hospital discharge and 1 year, but not at 2 years.¹²

Delirium

Delirium affects 60% to 80% of mechanically ventilated patients and often goes undiagnosed in the absence of systematic screening.^{110–113} It is an independent risk factor for ICU and hospital length of stay,^{113–115} increased cost,¹¹⁶ long-term neurocognitive impairment,^{117–118} and death.^{119–121} In addition, there are studies that link the disturbing dreams or persecutory delusions experienced as part of delirium as important risk factors for the long-term development of PTSD (Fig. 68-7).¹²²

Figure 68-7



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Picture drawn by a critically ill patient illustrating the violent and persecutory delusions experienced during active delirium while in the ICU. This picture illustrates how patients perceive threats to their safety and violent acts being perpetrated upon them during their stay in the ICU. (Used, with permission, from Griffiths and Jones.⁹)

There are emerging data highlighting the importance of delirium either as a marker for or part of the causal pathway for long-term neurocognitive dysfunction. A review of the association between delirium and long-term cognitive dysfunction¹⁰³ reported four studies with greater decline on neurocognitive measures at follow-up among patients experiencing delirium during hospitalization and four studies that found a higher incidence of dementia at long-term follow-up in elderly patients.¹⁰³ More recent studies in patients who had documented delirium have revealed long-term neurocognitive impairment in one-third of patients at 6-month follow-up.⁷ This relationship has also been observed at 3 and 12 months in a different patient sample, and neuropsychological impairment was also significantly linked to days of delirium.¹²³

Glucose Dysregulation

The relationship between blood glucose dysregulation and neurocognitive function has been evaluated at 1-year posthospital discharge,¹²⁴ and demonstrated an important relationship between moderate hyperglycemia during ICU and compromised neurocognitive outcomes in ARDS survivors. Recent data from surgical critically ill patients showed that hypoglycemia, hyperglycemia, and important fluctuations in blood glucose were also associated with compromise in cognitive outcome.¹²⁵ Hypoglycemia has been associated with a 3.6-fold increased risk of depression in patients with ARDS, further underscoring the relationship between glucose levels and neurological sequelae.¹²⁶

Sedatives or Analgesics

The impact of sedatives, narcotics, and anesthetics on long-term neurocognitive function is conflicting, although there may be an important early signal in reports suggesting that neurotoxic effects are more pronounced in high-risk groups, such as the elderly (>75 years) or those who have sustained recent brain injury or neurocognitive impairment.^{127,128} A recent substudy of the Awakening and Breathing Trial, however, showed that composite cognitive scores at 3-month and 12-month follow-up were similar between the intervention arm (spontaneous daily awakening and spontaneous breathing trials, with consequent less sedation exposure) compared to usual practice with spontaneous breathing trials.¹²⁹

Inflammation, Systemic Inflammatory Response Syndrome, and Sepsis

Neurocognitive morbidity may also be generated by inflammatory mediators and cytokines propagated by sepsis and the systemic inflammatory response syndrome. The brain is vulnerable to systemic inflammatory reactions in sepsis¹⁰⁴ that penetrate the blood-brain barrier and may modulate brain activity. Endotoxin-induced inflammation and cytokine activation are associated with impaired learning and memory,¹³⁰ and neurocognitive dysfunction.¹³¹ Animal models of sepsis demonstrate increased levels of interleukins 1 and 6 in the prefrontal cortex and hippocampus that are associated with learning and memory tasks.^{132–133} Congruent with this finding, a recent long-term outcome study showed that sepsis survivors had neurocognitive impairments several years after hospital discharge.¹³⁴

Caregiver and Family Burden in Critical Illness

The impact of a severe episode of critical illness on the caregiver and family has only recently gained momentum as an important issue in critical care medicine. There is no question that the caregiver suffers enormously during and after the illness. Caregivers conduct bedside vigils for their loved ones during the ICU stay, and then are saddled with virtually complete responsibility for the medical and psychological care of their family member who often has new complex medical needs, perhaps unresolved delirium, and new functional dependence. In some ways, it is surprising that it has taken so long for the caregiver issues to have been prioritized and acknowledged as a key component in the patient's rehabilitation program and ultimate outcome.

Recent work indicates that close to 60% of ICU survivors who received long-term mechanical ventilation still required the assistance of a family caregiver 1 year after their critical illness.²⁸ Existing evidence suggests that providing such care may have a deleterious impact on caregivers, and may compromise health-related quality of life compared with age-matched and sex-matched persons.¹⁸ There are also reports of important mental health issues in caregivers, including PTSD,¹³⁵ emotional distress,^{18,136–138} caregiver burden,¹³⁹ depression,¹⁴⁰ and anxiety.¹⁴¹

In a recent review, Johnson¹⁴⁴ concluded that caregivers experience burden because of the patient's physical and psychological dysfunction and the challenges of managing complex care in the home. Lifestyle disruption and provision of high levels of care¹⁹ also contribute to poor caregiver outcome.^{19,145} Much of the current research work has been limited by the cross-sectional design and follow-up to 1 year after hospital discharge. It is likely that caregiver needs will change over time as patients move through different transitions in their recovery¹⁴⁶ and attempt to return to work and resume their prior lifestyle.¹⁴⁷

Continuum of Care during and after Critical Illness: Intensive Care Unit and Post-Intensive Care Unit Rehabilitation

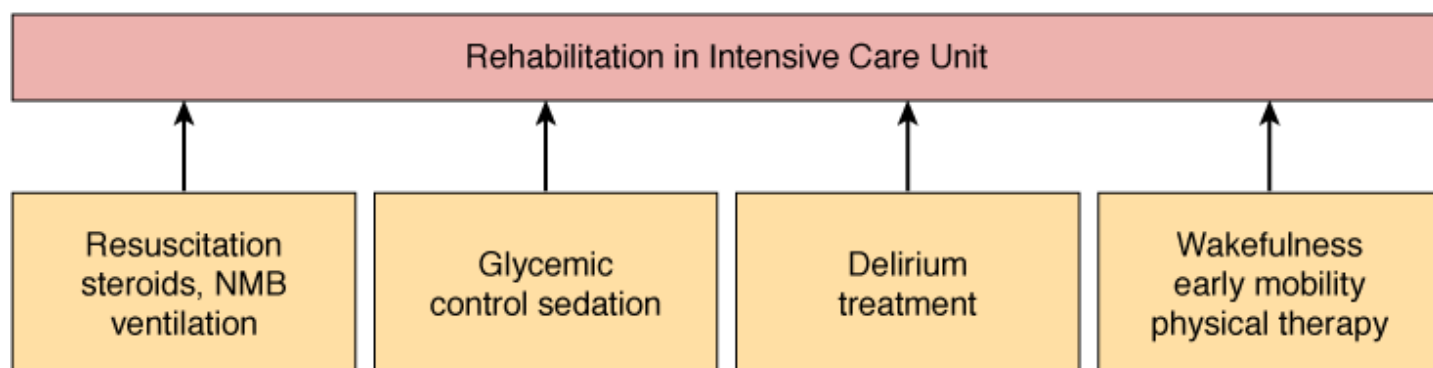
There are many barriers to improving outcome after critical illness in patients and family caregivers. First, we need to embrace the concept of a family-centered longitudinal model of care and recovery where we acknowledge that both patients and family caregivers have important needs during and after the ICU stay and that these needs will evolve and change at different transition points during recovery.¹⁴⁶ Second, we need to be honest and acknowledge that current ICU practice patterns may result in additional injury to vulnerable end organs, and that muscle, nerve, and brain need to be prioritized as key organ systems that largely determine functional, neurocognitive, and health-related quality-of-life outcomes in our patients. Third, we need to design studies that help elucidate the relationships between practice and long-term outcome so we can better determine which morbidities are potentially modifiable. Fourth, we need to understand how to risk stratify and optimize subsequent acute and chronic rehabilitative needs, functional independence, and return to work and/or previous lifestyle. Fifth, we have minimal understanding of the basic science of ICU-acquired morbidity, and this needs urgent attention. Finally, we need to embrace, empower, and educate each member of our interprofessional teams to help them understand that what we practice every day may change the long-term outcome for our patients and their families.

Figure 68-8 outlines the concept that rehabilitation begins the moment the patient is admitted to the ICU and that the way in which we conduct our treatment may have important downstream ramifications. The effectiveness of resuscitation affects mortality and propagation of organ dysfunction and perhaps this also has implications for muscle, nerve, and brain injury.¹⁴⁷ The diaphragm is very vulnerable to rest and undergoes rapid proteolysis within hours of controlled mechanical ventilation.³⁴ Attempting to prioritize spontaneous ventilation modes to preserve the diaphragm makes sense but there are no studies that link spontaneous

ventilation to improved longer-term outcome. The indication for and use of neuromuscular blockers and steroids remains controversial.

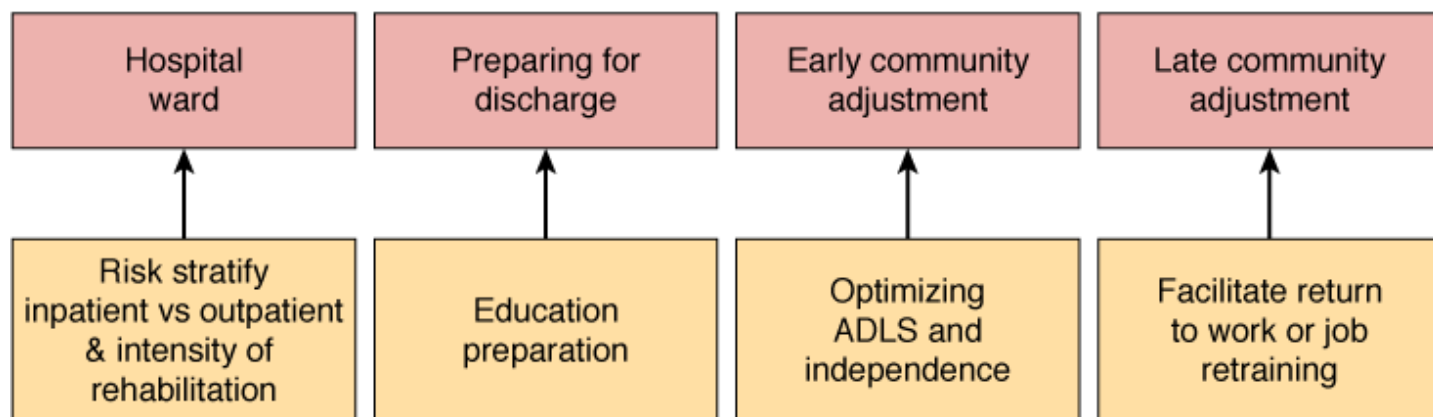
Figure 68-8

Phase-Specific Approach to Rehabilitation in Critical Illness: Targeting Muscle, Nerve, and Brain During and After the Intensive Care Unit Stay



Phase-Specific Approach to Rehabilitation in Critical Illness: Targeting Muscle, Nerve, and Brain During and After the Intensive Care Unit Stay

Rehabilitation after Intensive Care Unit—Phase-Specific Needs



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Rehabilitation continuum during and after the ICU stay. This highlights the concept that rehabilitation begins the moment the patient is admitted to the ICU and that our daily management may have an important impact on long-term outcome. It also highlights the various patient needs after they leave hospital and attempt reintegration into their prior lifestyle. (Adapted, with permission, from Herridge MS. Legacy of intensive care unit-acquired weakness. *Crit Care Med*. 2009;37[10 Suppl]:S457–S461.)

Perhaps one of the most fundamental limitations in constructing appropriate rehabilitation programs after critical illness is our current inability to risk stratify our patients. Risk stratification is a fundamental tenet upon which many other disciplines devise robust algorithmic treatment approaches to clinical problems. The heterogeneity of critically ill populations is an important barrier that needs to be addressed to be able to understand differences in functional outcome across different patient groups and the various factors that appear to drive a broad spectrum of outcome. For the sake of discussion and contrast, there is an enormous

difference in functional outcome between relatively young survivors of ARDS^{3,24} compared with older chronically critically ill patients,¹ despite the apparent similarities of severity of illness and a protracted ICU length of stay. The young, previously working, lung-injured group had few comorbid disorders, very low mortality after ICU discharge, and a significant, albeit less than predicted, improvement in functional status at 1 year, and virtually all returned to independent living. This is in stark contrast to an older group of chronically critically ill patients with a significant burden of comorbid disease, with a 44% mortality at 1-year after ICU discharge and only 11% achieving a good (alive with no functional dependency) outcome.

Current interventional work has focused on early mobility, which a number of studies has shown to be safe and feasible and alters short-term outcome in those patients who were previously functional.^{148–153} It is practical and logical to trial physiotherapy and occupational therapy interventions in those for whom there is a high likelihood for benefit.¹⁵² This approach, however, although important and laudable, will not determine how interventions should be tailored to meet individual needs nor differentially applied because there are almost no guidelines on specific patient subgroups. For example, offering such interventions to subpopulations of patients whose muscles and nerves have sustained such profound injury that they have lost any potential for rehabilitation raises inappropriate expectation. There are currently many proposed models for complex rehabilitation after critical illness, but none has focused on how to tailor the program to individual need nor shown long-term efficacy.^{154–157} A recent multicenter randomized trial of 286 critically ill patients assessed health-related quality of life and compared outcomes from a nurse-led intensive follow-up versus usual care at 12 months. There was no difference in health-related quality of life on the physical or mental health component scores, and the nurse-led follow-up program cost significantly more than usual care.¹⁵⁸ A self-help manual with instructions for physical therapy, however, improved 6-month outcomes in physical function assessed using the SF-36 health-related quality-of-life instrument; perhaps, patients and families could tailor this guide to individual need, although this consideration was not studied explicitly in this trial.¹⁵⁹

I have outlined important neuropsychological disabilities in this chapter, and there has been some early work evaluating potential interventions to improve these outcomes. Jones et al evaluated whether a prospectively collected diary of a patient's ICU stay could reduce the development of new onset PTSD during convalescence after critical illness.⁹⁴ Patients with an ICU stay of more than 72 hours were recruited to the study and intervention patients received their ICU diary at 1 month after ICU discharge, and assessment for the development of PTSD was made at 3 months. They found that use of the diary was associated with a decrease in new-onset PTSD. These early data are very promising but further understanding of the longer-term effect of the diary intervention is warranted.

Summary and Conclusion

The current state of the art in the outcomes literature suggests that patients will sustain some degree of neuromuscular, functional, and neuropsychological morbidity as a result of their critical illness, and this does not appear to be wholly reversible over time, even in younger patients who were previously working and highly functional. Pulmonary outcomes will likely be very good in those who are younger and have no premorbid

pulmonary disease. Health care costs after critical illness mirror those of chronic disease, but younger patients who receive support in transitioning back to work will overwhelmingly return to work. Patients with more comorbid illness and premorbid functional disability will have poor longer-term outcomes. Family caregivers may acquire new mood disorders that impair their health-related quality of life and may modify outcomes in patients. ICU-acquired weakness and brain injury represent major morbidities. Further basic science studies to elucidate the pathophysiology of brain, nerve, and muscle injury are urgently needed, as is work on risk stratification so that rehabilitation programs can be tailored to individual and family needs.

References

1. Unroe M, Kahn JM, Carson SS, et al. One-year trajectories of care and resource utilization for recipients of prolonged mechanical ventilation: a cohort study. *Ann Intern Med.* 2010;153(3):167–175. [[PubMed: 20679561](#)]

2. Needham DM, Dennison CR, Dowdy DW, et al. Study protocol: the Improving Care of Acute Lung Injury Patients (ICAP) study. *Crit Care.* 2006;10(1):R9.

3. Herridge MS, Cheung AM, Tansey CM, et al. One-year outcomes in survivors of the acute respiratory distress syndrome. *N Engl J Med.* 2003;348(8):683–693. [[PubMed: 12594312](#)]

4. Iwashyna TJ, Ely EW, Smith DM, Langa KM. Long-term cognitive impairment and functional disability among survivors of severe sepsis. *JAMA.* 2010;304(16):1787–1794. [[PubMed: 20978258](#)]

5. Herridge MS, Tansey CM, Matte A, et al. Functional disability five years after acute respiratory distress syndrome. *N Engl J Med.* 2011;364:1293–1304. [[PubMed: 21470008](#)]

6. Davydow DS, Desai SV, Needham DM, Bienvenu OJ. Psychiatric morbidity in survivors of the acute respiratory distress syndrome: a systematic review. *Psychosom Med.* 2008;70(4):512–519. [[PubMed: 18434495](#)]

7. Jackson JC, Hart RP, Gordon SM, et al. Six-month neuropsychological outcome of medical intensive care unit patients. *Crit Care Med.* 2003;31(4):1226–1234. [[PubMed: 12682497](#)]

8. Jones C, Griffiths RD, Slater T, et al. Significant cognitive dysfunction in non-delirious patients identified during and persisting following critical illness. *Intensive Care Med.* 2006;32(6):923–926. [[PubMed: 16525845](#)]

9. Griffiths RD, Jones C. Recovery from intensive care. *BMJ.* 1999; 319(7207):427–429. [[PubMed: 10445926](#)]

10. Angus DC, Musthafa AA, Clermont G, et al. Quality-adjusted survival in the first year after the acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2001;163(6):1389–1394. [[PubMed: 11371406](#)]

11. Hopkins RO, Weaver LK, Pope D, et al. Neuropsychological sequelae and impaired health status in survivors of severe acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;160(1):50–56. [[PubMed: 10390379](#)]

12. Hopkins RO, Weaver LK, Collingridge D, et al. Two-year cognitive, emotional, and quality-of-life outcomes in acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2005;171(4):340–347. [[PubMed: 15542793](#)]

13. Cheung AM, Tansey CM, Tomlinson G, et al. Two-year outcomes, health care use, and costs of survivors of acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2006;174(5):538–544. [[PubMed: 16763220](#)]

14. Angus DC, Clermont G, Linde-Zwirble WT, et al. Healthcare costs and long-term outcomes after acute respiratory distress syndrome: a phase III trial of inhaled **nitric oxide**. *Crit Care Med*. 2006;34(12):2883–2890. [[PubMed: 17075373](#)]

15. Pochard F, Darmon M, Fassier T, et al. Symptoms of anxiety and depression in family members of intensive care unit patients before discharge or death. A prospective multicenter study. *J Crit Care*. 2005;20(1):90–96. [[PubMed: 16015522](#)]

16. Azoulay E, Pochard F, Kentish-Barnes N, et al. Risk of post-traumatic stress symptoms in family members of intensive care unit patients. *Am J Respir Crit Care Med*. 2005;171(9):987–994. [[PubMed: 15665319](#)]

17. Van P, Schulz R, Chelluri L, Pinsky MR. Patient-specific, time-varying predictors of post-ICU informal caregiver burden: the caregiver outcomes after ICU discharge project. *Chest*. 2010;137(1):88–94.

18. Cameron JI, Herridge MS, Tansey CM, et al. Well-being in informal caregivers of survivors of acute respiratory distress syndrome. *Crit Care Med*. 2006;34(1):81–86. [[PubMed: 16374160](#)]

19. McHugh LG, Milberg JA, Whitcomb ME, et al. Recovery of function in survivors of the acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 1994;150(1):90–94. [[PubMed: 8025779](#)]

20. Weinert CR, Gross CR, Kangas JR, et al. Health-related quality of life after acute lung injury. *Am J Respir Crit Care Med*. 1997;156 (4 Pt 1):1120–1128.

21. Angus DC, Carlet J. Surviving intensive care: a report from the 2002 Brussels Roundtable. *Intensive Care Med*. 2003;29(3):368–377. [[PubMed: 12536269](#)]

22. Davidson TA, Caldwell ES, Curtis JR, et al. Reduced quality of life in survivors of acute respiratory distress syndrome compared with critically ill control patients. *JAMA*. 1999;281(4):354–360. [[PubMed: 9929089](#)]

23. Rubenfeld GD, Caldwell E, Peabody E, et al. Incidence and outcomes of acute lung injury. *N Engl J Med*. 2005;353(16):1685–1693. [[PubMed: 16236739](#)]

24. Dowdy DW, Eid MP, Dennison CR, et al. Quality of life after acute respiratory distress syndrome: a meta-analysis. *Intensive Care Med*. 2006;32(8):1115–1124. [[PubMed: 16783553](#)]

25. Ware JE Jr, Kosinski M. *SF-36 Physical and Mental Health Summary Scores: A Manual for Users of Version 1*. 2nd ed. New York: Quality Metrics; 2005.

-
26. Schelling G, Stoll C, Haller M, et al. Health-related quality of life and posttraumatic stress disorder in survivors of the acute respiratory distress syndrome. *Crit Care Med*. 1998;26(4):651–659. [[PubMed: 9559601](#)]
-
27. Orme J Jr, Romney JS, Hopkins RO, et al. Pulmonary function and health-related quality of life in survivors of acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2003;167(5):690–694. [[PubMed: 12493646](#)]
-
28. Chelluri L, Im KA, Belle SH, et al. Long-term mortality and quality of life after prolonged mechanical ventilation. *Crit Care Med*. 2004;32(1):61–69. [[PubMed: 14707560](#)]
-
29. Combes A, Costa MA, Trouillet JL, et al. Morbidity, mortality, and quality-of-life outcomes of patients requiring > or = 14 days of mechanical ventilation. *Crit Care Med*. 2003;31(5):1373–1381. [[PubMed: 12771605](#)]
-
30. Somme D, Maillet JM, Gisselbrecht M, et al. Critically ill old and the oldest-old patients in intensive care: short- and long-term outcomes. *Intensive Care Med*. 2003;29(12):2137–2143. [[PubMed: 14614546](#)]
-
31. Chelluri L, Pinsky MR, Donahoe MP, Grenvik A. Long-term outcome of critically ill elderly patients requiring intensive care. *JAMA*. 1993;269(24):3119–3123. [[PubMed: 8505814](#)]
-
32. Rockwood K, Noseworthy TW, Gibney RT, et al. One-year outcome of elderly and young patients admitted to intensive care units. *Crit Care Med*. 1993;21(5):687–691. [[PubMed: 8482089](#)]
-
33. Khouli H, Astua A, Dombrowski W, et al. Changes in health-related quality of life and factors predicting long-term outcomes in older adults admitted to intensive care units. *Crit Care Med*. 2011;39(4):731–737. [[PubMed: 21263318](#)]
-
34. Levine S, Nguyen T, Taylor N, et al. Rapid disuse atrophy of diaphragm fibers in mechanically ventilated humans. *N Engl J Med*. 2008;358(13):1327–1335. [[PubMed: 18367735](#)]
-
35. De Jonghe B, Sharshar T, Lefaucheur JP, et al. Paresis acquired in the intensive care unit: a prospective multicenter study. *JAMA*. 2002;288(22):2859–2867.
-
36. Leijten FS, Harinck-de Weerd JE, Poortvliet DC, de Weerd AW. The role of polyneuropathy in motor convalescence after prolonged mechanical ventilation. *JAMA*. 1995;274(15):1221–1225. [[PubMed: 7563512](#)]
-
37. Zifko UA. Long-term outcome of critical illness polyneuropathy. *Muscle Nerve Suppl*. 2000;9:S49–S52.
-
38. Fletcher SN, Kennedy DD, Ghosh IR, et al. Persistent neuromuscular and neurophysiologic abnormalities in long-term survivors of prolonged critical illness. *Crit Care Med*. 2003;31(4):1012–1016. [[PubMed: 12682465](#)]
-
39. Kerbaul F, Brousse M, Collart F, et al. Combination of histopathological and electromyographic patterns can help to evaluate functional outcome of critical ill patients with neuromuscular weakness syndromes. *Crit Care*. 2004;8(6):R358–R366.
-

40. Guarneri B, Bertolini G, Latronico N. Long-term outcome in patients with critical illness myopathy or neuropathy: the Italian multicentre CRIMYNE study. *J Neurol Neurosurg Psychiatry*. 2008;79(7):838–841. [\[PubMed: 18339730\]](#)

41. de Seze M, Petit H, Wiart L, et al. Critical illness polyneuropathy. A 2-year follow-up study in 19 severe cases. *Eur Neurol*. 2000;43(2):61–69.

42. De Jonghe B, Bastuji-Garin S, Durand MC, et al. Respiratory weakness is associated with limb weakness and delayed weaning in critical illness. *Crit Care Med*. 2007;35(9):2007–2015.

43. Garnacho-Montero J, Maya-Villar R, Garcia-Garmendia JL, et al. Effect of critical illness polyneuropathy on the withdrawal from mechanical ventilation and the length of stay in septic patients. *Crit Care Med*. 2005;33(2):349–354. [\[PubMed: 15699838\]](#)

44. Bercker S, Weber-Carstens S, Deja M, et al. Critical illness polyneuropathy and myopathy in patients with acute respiratory distress syndrome. *Crit Care Med*. 2005;33(4):711–715. [\[PubMed: 15818093\]](#)

45. Rudis MI, Guslits BJ, Peterson EL, et al. Economic impact of prolonged motor weakness complicating neuromuscular blockade in the intensive care unit. *Crit Care Med*. 1996;24(10):1749–1756. [\[PubMed: 8874316\]](#)

46. Ali NA, O'Brien JM Jr, Hoffmann SP, et al. Acquired weakness, handgrip strength, and mortality in critically ill patients. *Am J Respir Crit Care Med*. 2008;178(3):261–268. [\[PubMed: 18511703\]](#)

47. Stevens RD, Marshall SA, Cornblath DR, et al. A framework for diagnosing and classifying intensive care unit-acquired weakness. *Crit Care Med*. 2009;37(10 Suppl):S299–S308.

48. Bolton CF, Gilbert JJ, Hahn AF, Sibbald WJ. Polyneuropathy in critically ill patients. *J Neurol Neurosurg Psychiatry*. 1984;47(11):1223–1231. [\[PubMed: 6094735\]](#)

49. de Letter MA, Schmitz PI, Visser LH, et al. Risk factors for the development of polyneuropathy and myopathy in critically ill patients. *Crit Care Med*. 2001;29(12):2281–2286.

50. Stevens RD, Dowdy DW, Michaels RK, et al. Neuromuscular dysfunction acquired in critical illness: a systematic review. *Intensive Care Med*. 2007;33(11):1876–1891. [\[PubMed: 17639340\]](#)

51. Fenzi F, Latronico N, Refatti N, Rizzuto N. Enhanced expression of E-selectin on the vascular endothelium of peripheral nerve in critically ill patients with neuromuscular disorders. *Acta Neuropathol*. 2003;106(1):75–82. [\[PubMed: 12698264\]](#)

52. Hotchkiss RS, Swanson PE, Freeman BD, et al. Apoptotic cell death in patients with sepsis, shock, and multiple organ dysfunction. *Crit Care Med*. 1999;27(7):1230–1251. [\[PubMed: 10446814\]](#)

53. van den Berghe G, Wouters P, Weekers F, et al. Intensive [insulin](#) therapy in the critically ill patients. *N Engl J Med*. 2001;345(19):1359–1367.
-
54. van den Berghe G, Wilmer A, Hermans G, et al. Intensive [insulin](#) therapy in the medical ICU. *N Engl J Med*. 2006;354(5):449–461.
-
55. Vanhorebeek I, De Vos R, Mesotten D, et al. Protection of hepatocyte mitochondrial ultrastructure and function by strict blood glucose control with [insulin](#) in critically ill patients. *Lancet*. 2005;365(9453):53–59. [[PubMed: 15639679](#)]
-
56. Deem S, Lee CM, Curtis JR. Acquired neuromuscular disorders in the intensive care unit. *Am J Respir Crit Care Med*. 2003;168(7):735–739. [[PubMed: 14522811](#)]
-
57. Siroen MP, van Leeuwen PA, Nijveldt RJ, et al. Modulation of asymmetric dimethylarginine in critically ill patients receiving intensive [insulin](#) treatment: a possible explanation of reduced morbidity and mortality? *Crit Care Med*. 2005;33(3):504–510. [[PubMed: 15753739](#)]
-
58. Aljada A, Ghanim H, Mohanty P, et al. [Insulin](#) inhibits the pro-inflammatory transcription factor early growth response gene-1 (Egr)-1 expression in mononuclear cells (MNC) and reduces plasma tissue factor (TF) and plasminogen activator inhibitor-1 (PAI-1) concentrations. *J Clin Endocrinol Metab*. 2002;87(3):1419–1422. [[PubMed: 11889219](#)]
-
59. Ishii DN, Lupien SB. Insulin-like growth factors protect against diabetic neuropathy: effects on sensory nerve regeneration in rats. *J Neurosci Res*. 1995;40(1):138–144. [[PubMed: 7714922](#)]
-
60. Garnacho-Montero J, Madrazo-Osuna J, Garcia-Garmendia JL, et al. Critical illness polyneuropathy: risk factors and clinical consequences. A cohort study in septic patients. *Intensive Care Med*. 2001;27(8):1288–1296. [[PubMed: 11511941](#)]
-
61. Leatherman JW, Fluegel WL, David WS, et al. Muscle weakness in mechanically ventilated patients with severe asthma. *Am J Respir Crit Care Med*. 1996;153(5):1686–1690. [[PubMed: 8630621](#)]
-
62. Behbehani NA, Al-Mane F, D'Yachkova Y, et al. Myopathy following mechanical ventilation for acute severe asthma: the role of muscle relaxants and corticosteroids. *Chest*. 1999;115(6):1627–1631. [[PubMed: 10378560](#)]
-
63. MacFarlane IA, Rosenthal FD. Severe myopathy after status asthmaticus. *Lancet*. 1977;2(8038):615. [[PubMed: 71437](#)]
-
64. Thiele RI, Jakob H, Hund E, et al. Sepsis and catecholamine support are the major risk factors for critical illness polyneuropathy after open heart surgery. *Thorac Cardiovasc Surg*. 2000;48(3):145–150. [[PubMed: 10903060](#)]
-

65. Pandit L, Agrawal A. Neuromuscular disorders in critical illness. *Clin Neurol Neurosurg*. 2006;108(7):621–627. [[PubMed: 16730883](#)]
-
66. Papazian L, Forel JM, Gacouin A, et al. Neuromuscular blockers in early acute respiratory distress syndrome. *N Engl J Med*. 2010;363(12):1107–1116. [[PubMed: 20843245](#)]
-
67. Latronico N, Fenzi F, Recupero D, et al. Critical illness myopathy and neuropathy. *Lancet*. 1996;347(9015):1579–1582. [[PubMed: 8667865](#)]
-
68. Campellone JV, Lacomis D, Kramer DJ, et al. Acute myopathy after liver transplantation. *Neurology*. 1998;50(1):46–53. [[PubMed: 9443456](#)]
-
69. Helliwell TR, Coakley JH, Wagenmakers AJ, et al. Necrotizing myopathy in critically-ill patients. *J Pathol*. 1991;164(4):307–314. [[PubMed: 1919868](#)]
-
70. Lacomis D. Critical illness myopathy. *Curr Rheumatol Rep*. 2002; 4(5):403–408. [[PubMed: 12217245](#)]
-
71. Winkelman C. Inactivity and inflammation: selected cytokines as biologic mediators in muscle dysfunction during critical illness. *AACN Clin Issues*. 2004;15(1):74–82. [[PubMed: 14767366](#)]
-
72. Allen DC, Arunachalam R, Mills KR. Critical illness myopathy: further evidence from muscle-fiber excitability studies of an acquired channelopathy. *Muscle Nerve*. 2008;37(1):14–22. [[PubMed: 17763454](#)]
-
73. Neff TA, Stocker R, Frey HR, et al. Long-term assessment of lung function in survivors of severe ARDS. *Chest*. 2003;123(3):845–853. [[PubMed: 12628887](#)]
-
74. Schelling G, Stoll C, Vogelmeier C, et al. Pulmonary function and health-related quality of life in a sample of long-term survivors of the acute respiratory distress syndrome. *Intensive Care Med*. 2000; 26(9):1304–1311. [[PubMed: 11089757](#)]
-
75. Nobauer-Huhmann IM, Eibenberger K, Schaefer-Prokop C, et al. Changes in lung parenchyma after acute respiratory distress syndrome (ARDS): assessment with high-resolution computed tomography. *Eur Radiol*. 2001;11(12):2436–2443. [[PubMed: 11734936](#)]
-
76. Desai SR, Wells AU, Rubens MB, et al. Acute respiratory distress syndrome: CT abnormalities at long-term follow-up. *Radiology*. 1999;210(1):29–35. [[PubMed: 9885583](#)]
-
77. Linden VB, Lidegran MK, Frisen G, et al. ECMO in ARDS: a long-term follow-up study regarding pulmonary morphology and function and health-related quality of life. *Acta Anaesthesiol Scand*. 2009;53(4):489–495. [[PubMed: 19226296](#)]
-
78. Hudson SJ, Brett SJ. Heterotopic ossification—a long-term consequence of prolonged immobility. *Crit Care*. 2006;10(6):174. [[PubMed: 17129365](#)]

-
79. Adhikari NK, McAndrews MP, Tansey CM, et al. Self-reported symptoms of depression and memory dysfunction in survivors of ARDS. *Chest*. 2009;135(3):678–687. [[PubMed: 19265087](#)]
-
80. Mikkelsen ME, Shull WH, Biester RC, et al. Cognitive, mood and quality of life impairments in a select population of ARDS survivors. *Respirology*. 2009;14(1):76–82. [[PubMed: 19144052](#)]
-
81. Davydow DS, Gifford JM, Desai SV, et al. Depression in general intensive care unit survivors: a systematic review. *Intensive Care Med*. 2009;35(5):796–809. [[PubMed: 19165464](#)]
-
82. Hopkins RO, Key CW, Suchyta MR, et al. Risk factors for depression and anxiety in survivors of acute respiratory distress syndrome. *Gen Hosp Psychiatry*. 2010;32(2):147–155. [[PubMed: 20302988](#)]
-
83. Dowdy DW, Bienvenu OJ, Dinglas VD, et al. Are intensive care factors associated with depressive symptoms 6 months after acute lung injury? *Crit Care Med*. 2009;37(5):1702–1707. [[PubMed: 19357507](#)]
-
84. Horowitz M, Wilner N, Alvarez W. Impact of event scale: a measure of subjective stress. *Psychosom Med*. 1979;41(3):209–218. [[PubMed: 472086](#)]
-
85. Kapfhammer HP, Rothenhausler HB, Krauseneck T, et al. Posttraumatic stress disorder and health-related quality of life in long-term survivors of acute respiratory distress syndrome. *Am J Psychiatry*. 2004; 161(1):45–52. [[PubMed: 14702249](#)]
-
86. Davydow DS, Gifford JM, Desai SV, et al. Posttraumatic stress disorder in general intensive care unit survivors: a systematic review. *Gen Hosp Psychiatry*. 2008;30(5):421–434. [[PubMed: 18774425](#)]
-
87. Kapfhammer HP, Rothenhausler HB, Krauseneck T, et al. Posttraumatic stress disorder and health-related quality of life in long-term survivors of acute respiratory distress syndrome. *Am J Psychiatry*. 2004; 161(1):45–52. [[PubMed: 14702249](#)]
-
88. McCartney JR, Boland RJ. Anxiety and delirium in the intensive care unit. *Crit Care Clin*. 1994;10(4):673–680. [[PubMed: 8000920](#)]
-
89. Skodol AE. Anxiety in the medically ill: nosology and principles of differential diagnosis. *Semin Clin Neuropsychiatry*. 1999;4(2):64–71. [[PubMed: 10378950](#)]
-
90. Szokol JW, Vender JS. Anxiety, delirium, and pain in the intensive care unit. *Crit Care Clin*. 2001;17(4):821–842. [[PubMed: 11762263](#)]
-
91. Kiekkas P, Theodorakopoulou G, Spyratos F, Baltopoulos GI. Psychological distress and delusional memories after critical care: a literature review. *Int Nurs Rev*. 2010;57(3):288–296. [[PubMed: 20796057](#)]
-

92. Myhren H, Ekeberg O, Toien K, et al. Posttraumatic stress, anxiety and depression symptoms in patients during the first year post intensive care unit discharge. *Crit Care*. 2010;14(1):R14.
-
93. Bienvenu OJ, Neufeld KJ. Post-traumatic stress disorder in medical settings: focus on the critically ill. *Curr Psychiatry Rep*. 2011;13(1):3–9. [PubMed: 20981585]
-
94. Jones C, Backman C, Capuzzo M, et al. Intensive care diaries reduce new onset post traumatic stress disorder following critical illness: a randomised, controlled trial. *Crit Care*. 2010;14(5):R168.
-
95. Sukantarat KT, Burgess PW, Williamson RC, Brett SJ. Prolonged cognitive dysfunction in survivors of critical illness. *Anaesthesia*. 2005;60(9):847–853. [PubMed: 16115244]
-
96. Rothenhausler HB, Ehrentraut S, Stoll C, et al. The relationship between cognitive performance and employment and health status in long-term survivors of the acute respiratory distress syndrome: results of an exploratory study. *Gen Hosp Psychiatry*. 2001;23(2):90–96. [PubMed: 11313077]
-
97. Christie JD, Biester RC, Taichman DB, et al. Formation and validation of a telephone battery to assess cognitive function in acute respiratory distress syndrome survivors. *J Crit Care*. 2006;21(2):125–132. [PubMed: 16769455]
-
98. Duning T, Ellger B. Is hypoglycaemia dangerous? *Best Pract Res Clin Anaesthesiol*. 2009;23(4):473–485. [PubMed: 20108586]
-
99. Hopkins RO, Weaver LK, Chan KJ, Orme JF Jr. Quality of life, emotional, and cognitive function following acute respiratory distress syndrome. *J Int Neuropsychol Soc*. 2004;10(7):1005–1017. [PubMed: 15803563]
-
100. van der Schaaf M, Dettling DS, Beelen A, et al. Poor functional status immediately after discharge from an intensive care unit. *Disabil Rehabil*. 2008;30(23):1812–1818.
-
101. Ehlenbach WJ, Hough CL, Crane PK, et al. Association between acute care and critical illness hospitalization and cognitive function in older adults. *JAMA*. 2010;303(8):763–770. [PubMed: 20179286]
-
102. Starr JM, Whalley LJ. Drug-induced dementia. Incidence, management and prevention. *Drug Saf*. 1994;11(5):310–317. [PubMed: 7873091]
-
103. Jackson JC, Gordon SM, Hart RP, et al. The association between delirium and cognitive decline: a review of the empirical literature. *Neuropsychol Rev*. 2004;14(2):87–98. [PubMed: 15264710]
-
104. Elenkov IJ, Iezzoni DG, Daly A, et al. Cytokine dysregulation, inflammation and well-being. *Neuroimmunomodulation*. 2005;12(5):255–269. [PubMed: 16166805]
-
105. Hopkins RO, Myers CE, Shohamy D, et al. Impaired probabilistic category learning in hypoxic subjects with hippocampal damage. *Neuropsychologia*. 2004;42(4):524–535. [PubMed: 14728924]

-
106. Gale SD, Hopkins RO. Effects of hypoxia on the brain: neuroimaging and neuropsychological findings following carbon monoxide poisoning and obstructive sleep apnea. *J Int Neuropsychol Soc.* 2004;10(1):60–71. [\[PubMed: 14751008\]](#)
-
107. Hopkins RO, Gale SD, Johnson SC, et al. Severe anoxia with and without concomitant brain atrophy and neuropsychological impairments. *J Int Neuropsychol Soc.* 1995;1(5):501–509. [\[PubMed: 9375235\]](#)
-
108. Graham DI, Gennarelli TA, McIntosh TK. Trauma. In: Graham DI, Lantos PI, eds. *Greenfield's Neuropathology*. 7th ed. London, UK: Edward Arnold; 2002:823–882.
-
109. Oniki TA, Gardner RM. Computerized detection of arterial [oxygen](#) desaturations in an intensive care unit. *Proc Annu Symp Comput Appl Med Care.* 1993;356–360.
-
110. Ely EW, Inouye SK, Bernard GR, et al. Delirium in mechanically ventilated patients: validity and reliability of the confusion assessment method for the intensive care unit (CAM-ICU). *JAMA.* 2001;286(21):2703–2710. [\[PubMed: 11730446\]](#)
-
111. Spronk PE, Riekerk B, Hofhuis J, Rommes JH. Occurrence of delirium is severely underestimated in the ICU during daily care. *Intensive Care Med.* 2009;35(7):1276–1280. [\[PubMed: 19350214\]](#)
-
112. van Eijk MM, van Marum RJ, Klijn IA, et al. Comparison of delirium assessment tools in a mixed intensive care unit. *Crit Care Med.* 2009;37(6):1881–1885.
-
113. Ouimet S, Kavanagh BP, Gottfried SB, Skrobik Y. Incidence, risk factors and consequences of ICU delirium. *Intensive Care Med.* 2007;33(1):66–73. [\[PubMed: 17102966\]](#)
-
114. Thomason JW, Shintani A, Peterson JF, et al. Intensive care unit delirium is an independent predictor of longer hospital stay: a prospective analysis of 261 non-ventilated patients. *Crit Care.* 2005; 9(4):R375–R381.
-
115. Ely EW, Gautam S, Margolin R, et al. The impact of delirium in the intensive care unit on hospital length of stay. *Intensive Care Med.* 2001;27(12):1892–1900. [\[PubMed: 11797025\]](#)
-
116. Milbrandt EB, Deppen S, Harrison PL, et al. Costs associated with delirium in mechanically ventilated patients. *Crit Care Med.* 2004;32(4):955–962. [\[PubMed: 15071384\]](#)
-
117. Girard TD, Jackson JC, Pandharipande PP, et al. Delirium as a predictor of long-term cognitive impairment in survivors of critical illness. *Crit Care Med.* 2010;38(7):1513–1520. [\[PubMed: 20473145\]](#)
-
118. Jackson JC, Girard TD, Gordon SM, et al. Long-term cognitive and psychological outcomes in the awakening and breathing controlled trial. *Am J Respir Crit Care Med.* 2010;182(2):183–191. [\[PubMed: 20299535\]](#)
-

119. Ely EW, Shintani A, Truman B, et al. Delirium as a predictor of mortality in mechanically ventilated patients in the intensive care unit. *JAMA*. 2004;291(14):1753–1762. [[PubMed: 15082703](#)]
-
120. Pisani MA, Kong SY, Kasl SV, et al. Days of delirium are associated with 1-year mortality in an older intensive care unit population. *Am J Respir Crit Care Med*. 2009;180(11):1092–1097. [[PubMed: 19745202](#)]
-
121. Shehabi Y, Riker RR, Bokesch PM, et al. Delirium duration and mortality in lightly sedated, mechanically ventilated intensive care patients. *Crit Care Med*. 2010;38(12):2311–2318. [[PubMed: 20838332](#)]
-
122. Jones C, Backman C, Capuzzo M, et al. Precipitants of post-traumatic stress disorder following intensive care: a hypothesis generating study of diversity in care. *Intensive Care Med*. 2007;33(6):978–985. [[PubMed: 17384929](#)]
-
123. Girard TD, Jackson JC, Pandharipande PP, et al. Delirium as a predictor of long-term cognitive impairment in survivors of critical illness. *Crit Care Med*. 2010;38(7):1513–1520. [[PubMed: 20473145](#)]
-
124. Hopkins RO, Suchyta MR, Snow GL, et al. Blood glucose dysregulation and cognitive outcome in ARDS survivors. *Brain Inj*. 2010; 24(12):1478–1484. [[PubMed: 20858026](#)]
-
125. Duning T, van den Heuvel I, Dickmann A, et al. Hypoglycemia aggravates critical illness-induced neurocognitive dysfunction. *Diabetes Care*. 2010;33(3):639–644. [[PubMed: 20032274](#)]
-
126. Dowdy DW, Dinglas V, Mendez-Tellez PA, et al. Intensive care unit hypoglycemia predicts depression during early recovery from acute lung injury. *Crit Care Med*. 2008;36(10):2726–2733. [[PubMed: 18766087](#)]
-
127. Ancelin ML, de Roquefeuil G, Ledesert B, et al. Exposure to anaesthetic agents, cognitive functioning and depressive symptomatology in the elderly. *Br J Psychiatry*. 2001;178:360–366. [[PubMed: 11282816](#)]
-
128. Rasmussen LS, Johnson T, Kuipers HM, et al. Does anaesthesia cause postoperative cognitive dysfunction? A randomised study of regional versus general anaesthesia in 438 elderly patients. *Acta Anaesthesiol Scand*. 2003;47(3):260–266. [[PubMed: 12648190](#)]
-
129. Jackson JC, Girard TD, Gordon SM, et al. Long-term cognitive and psychological outcomes in the awakening and breathing controlled trial. *Am J Respir Crit Care Med*. 2010;182(2):183–191. [[PubMed: 20299535](#)]
-
130. Reichenberg A, Yirmiya R, Schuld A, et al. Cytokine-associated emotional and cognitive disturbances in humans. *Arch Gen Psychiatry*. 2001;58(5):445–452. [[PubMed: 11343523](#)]
-
131. Capuron L, Lamarque D, Dantzer R, Goodall G. Attentional and mnemonic deficits associated with infectious disease in humans. *Psychol Med*. 1999;29(2):291–297. [[PubMed: 10218921](#)]
-

132. Sparkman NL, Buchanan JB, Heyen JR, et al. Interleukin-6 facilitates lipopolysaccharide-induced disruption in working memory and expression of other proinflammatory cytokines in hippocampal neuronal cell layers. *J Neurosci*. 2006;26(42):10709–10716. [[PubMed: 17050710](#)]

133. Barichello T, Martins MR, Reinke A, et al. Cognitive impairment in sepsis survivors from cecal ligation and perforation. *Crit Care Med*. 2005;33(1):221–223; discussion 262–263. [[PubMed: 15644673](#)]

134. Rosene KA, Copass MK, Kastner LS, et al. Persistent neuropsychological sequelae of toxic shock syndrome. *Ann Intern Med*. 1982;96 (6 Pt 2):865–870.

135. Azoulay E, Pochard F, Kentish-Barnes N, et al. Risk of post-traumatic stress symptoms in family members of intensive care unit patients. *Am J Respir Crit Care Med*. 2005;171(9):987–994. [[PubMed: 15665319](#)]

136. Im K, Belle SH, Schulz R, et al. Prevalence and outcomes of caregiving after prolonged (> or = 48 hours) mechanical ventilation in the ICU. *Chest*. 2004;125(2):597–606. [[PubMed: 14769744](#)]

137. Jones C, Skirrow P, Griffiths RD, et al. Post-traumatic stress disorder-related symptoms in relatives of patients following intensive care. *Intensive Care Med*. 2004;30(3):456–460. [[PubMed: 14767589](#)]

138. Van Pelt DC, Milbrandt EB, Qin L, et al. Informal caregiver burden among survivors of prolonged mechanical ventilation. *Am J Respir Crit Care Med*. 2007;175(2):167–173.

139. Foster M, Chaboyer W. Family carers of ICU survivors: a survey of the burden they experience. *Scand J Caring Sci*. 2003;17(3):205–214. [[PubMed: 12919454](#)]

140. Douglas SL, Daly BJ, Kelley CG, et al. Impact of a disease management program upon caregivers of chronically critically ill patients. *Chest*. 2005;128(6):3925–3936. [[PubMed: 16354865](#)]

141. Pochard F, Darmon M, Fassier T, et al. Symptoms of anxiety and depression in family members of intensive care unit patients before discharge or death. A prospective multicenter study. *J Crit Care*. 2005;20(1):90–96. [[PubMed: 16015522](#)]

142. Johnson P, Chaboyer W, Foster M, van der Vooren R. Caregivers of ICU patients discharged home: what burden do they face? *Intensive Crit Care Nurs*. 2001;17(4):219–227. [[PubMed: 11868730](#)]

143. Van Pelt DC, Milbrandt EB, Qin L, et al. Informal caregiver burden among survivors of prolonged mechanical ventilation. *Am J Respir Crit Care Med*. 2007;175(2):167–173.

144. Lee CM, Herridge MS, Matte A, Cameron JI. Education and support needs during recovery in acute respiratory distress syndrome survivors. *Crit Care*. 2009;13(5):R153.

145. Cox CE, Docherty SL, Brandon DH, et al. Surviving critical illness: acute respiratory distress syndrome as experienced by patients and their caregivers. *Crit Care Med*. 2009;37(10):2702–2708. [[PubMed: 19865004](#)]

-
146. Cameron JI, Gignac MA. "Timing it right": a conceptual framework for addressing family caregivers' support needs from the hospital to the home. *Patient Educ Couns*. 2008;70(3):305–314. [PubMed: 18155388]
-
147. Rivers E, Nguyen B, Havstad S, et al. Early goal-directed therapy in the treatment of severe sepsis and septic shock. *N Engl J Med*. 2001;345(19):1368–1377. [PubMed: 11794169]
-
148. Needham DM. Mobilizing patients in the intensive care unit: improving neuromuscular weakness and physical function. *JAMA*. 2008;300(14):1685–1690. [PubMed: 18840842]
-
149. Bailey P, Thomsen GE, Spuhler VJ, et al. Early activity is feasible and safe in respiratory failure patients. *Crit Care Med*. 2007; 35(1):139–145. [PubMed: 17133183]
-
150. Morris PE, Goad A, Thompson C, et al. Early intensive care unit mobility therapy in the treatment of acute respiratory failure. *Crit Care Med*. 2008;36(8):2238–2243. [PubMed: 18596631]
-
151. Hopkins RO, Spuhler VJ, Thomsen GE. Transforming ICU culture to facilitate early mobility. *Crit Care Clin*. 2007;23(1):81–96. [PubMed: 17307118]
-
152. Schweickert WD, Pohlman MC, Pohlman AS, et al. Early physical and occupational therapy in mechanically ventilated, critically ill patients: a randomised controlled trial. *Lancet*. 2009;373(9678):1874–1882. [PubMed: 19446324]
-
153. Burtin C, Clerckx B, Robbeets C, et al. Early exercise in critically ill patients enhances short-term functional recovery. *Crit Care Med*. 2009;37(9):2499–2505. [PubMed: 19623052]
-
154. Rubenfeld GD. Post-hospital case management to improve clinical outcomes in individuals requiring mechanical ventilation. <http://clinicaltrials.gov/ct2/show/NCT00149513>.
-
155. Daly BJ, Douglas SL, Kelley CG, et al. Trial of a disease management program to reduce hospital readmissions of the chronically critically ill. *Chest*. 2005;128(2):507–517. [PubMed: 16100132]
-
156. Gosselink R, Bott J, Johnson M, et al. Physiotherapy for adult patients with critical illness: recommendations of the European Respiratory Society and European Society of Intensive Care Medicine Task Force on Physiotherapy for Critically Ill Patients. *Intensive Care Med*. 2008;34(7):1188–1199. [PubMed: 18283429]
-
157. Tan T, Brett SJ, Stokes T. Rehabilitation after critical illness: summary of NICE guidance. *BMJ*. 2009;338:b822.
-
158. Cuthbertson BH, Rattray J, Campbell MK, et al. The PRACTICaL study of nurse led, intensive care follow-up programmes for improving long term outcomes from critical illness: a pragmatic randomised controlled trial. *BMJ*. 2009;339:b3723.

159. Jones C, Skirrow P, Griffiths RD, et al. Rehabilitation after critical illness: a randomized, controlled trial. *Crit Care Med.* 2003;31(10):2456–2461. [[PubMed: 14530751](#)]

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